

# Part V

# Multi-Rater Modeling

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# Multi-Rater Modeling - Contents

1. Motivation: What is Disagreement
2. Ambiguous Ground-Truth, Imperfect Annotators
3. Evaluation and Multiple Raters
4. Hands-On

# 1. Motivation - What is Disagreement

Uncertain data creates predictive uncertainty (aleatoric)

If data were perfect, there would be no uncertainty

But what is data?

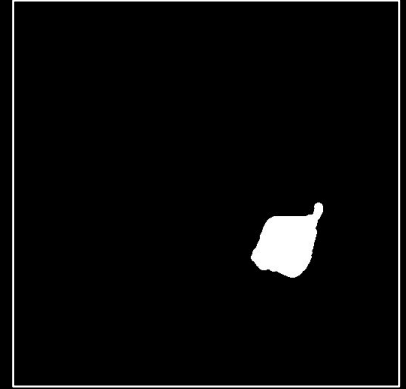


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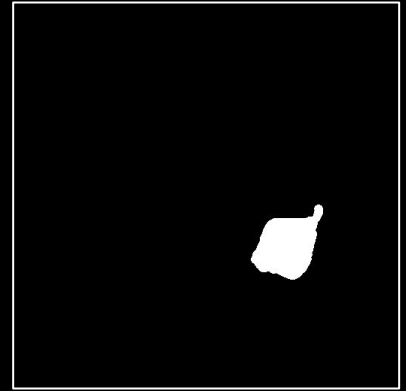
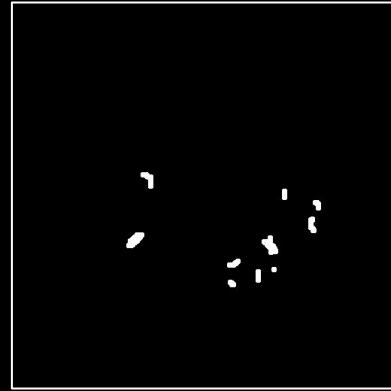


# 1. Motivation - What is Disagreement

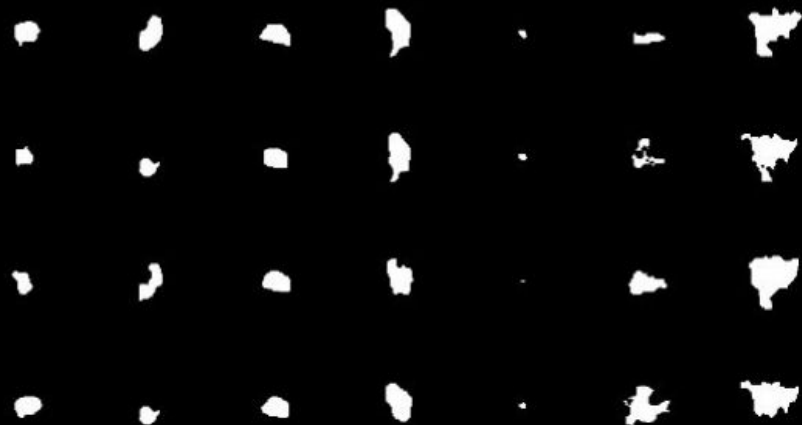
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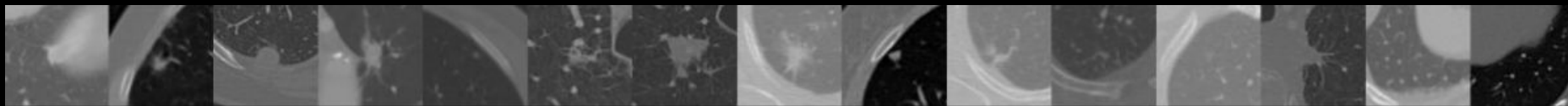
# 1. Motivation - What is Disagreement



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These may all be “valid”

# 1. Motivation - What is Disagreement

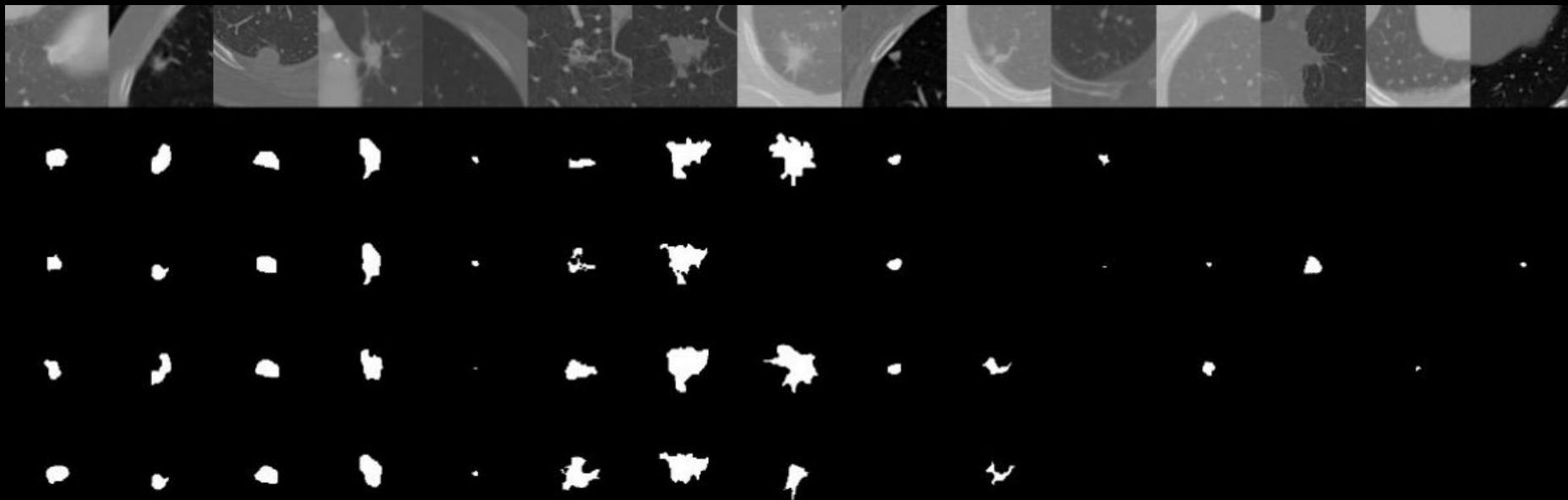


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These may all be “valid”

Something different is going on here

# 1. Motivation - What is Disagreement



We must not model these two scenarios in the same way

Note: we focus in segmentation here. A natural transition to a classification mindset is: *would this annotator think the same about this sample next week?*

## 2. Ambiguous Ground-Truth, Imperfect Annotators



## 2. Ambiguous Ground-Truth, Imperfect Annotators



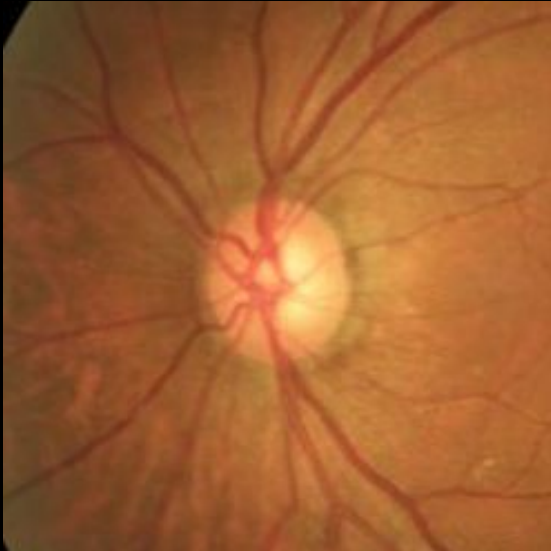
## 2. Ambiguous Ground-Truth, Imperfect Annotators

|                                                                                   |  |  |
|-----------------------------------------------------------------------------------|--|--|
|  |  |  |
|                                                                                   |  |  |
|                                                                                   |  |  |



## 2. Ambiguous Ground-Truth, Imperfect Annotators

|                                                                                   |                         |  |
|-----------------------------------------------------------------------------------|-------------------------|--|
|  | Exchangeable annotators |  |
|                                                                                   |                         |  |
|                                                                                   |                         |  |



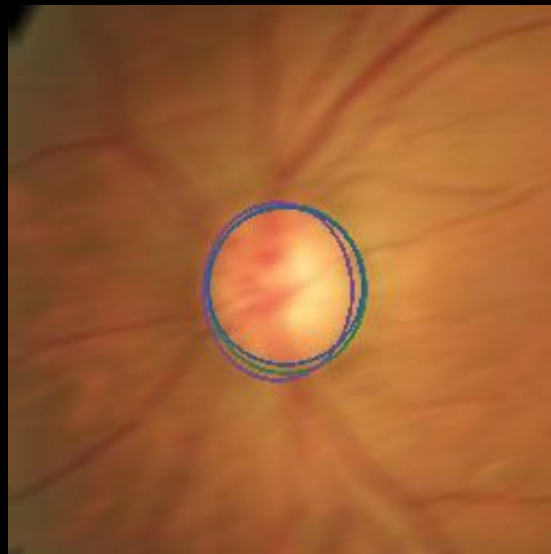
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|                                                                                   |                         |  |
|-----------------------------------------------------------------------------------|-------------------------|--|
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| $\nexists ! y^*$                                                                  |                         |  |
|                                                                                   |                         |  |



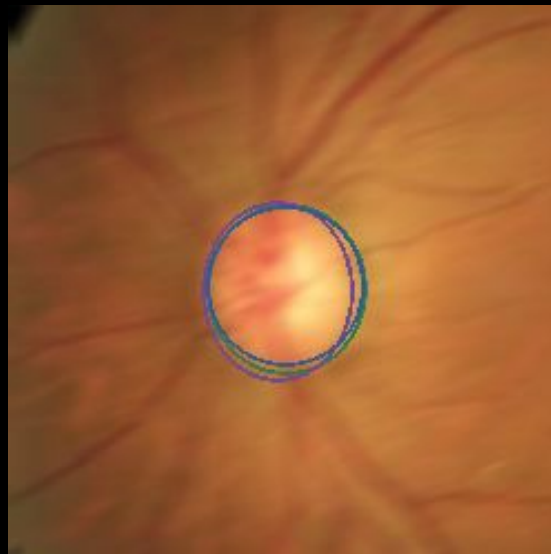
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|                                                                                   |                         |  |



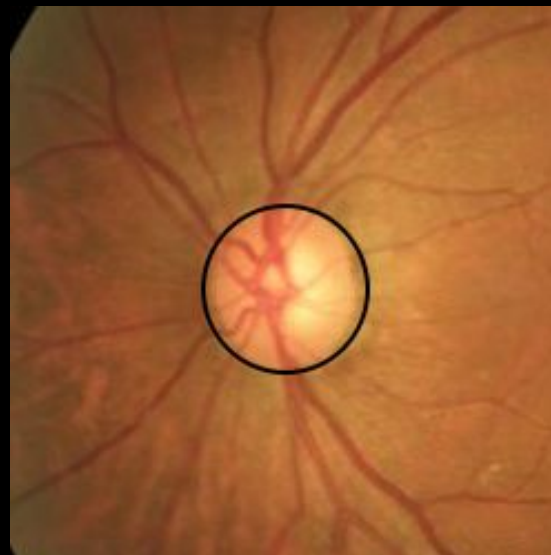
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|                                                                                   |                             |  |
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| $\nexists ! y^*$                                                                  | Aleatoric Image Uncertainty |  |
|                                                                                   |                             |  |



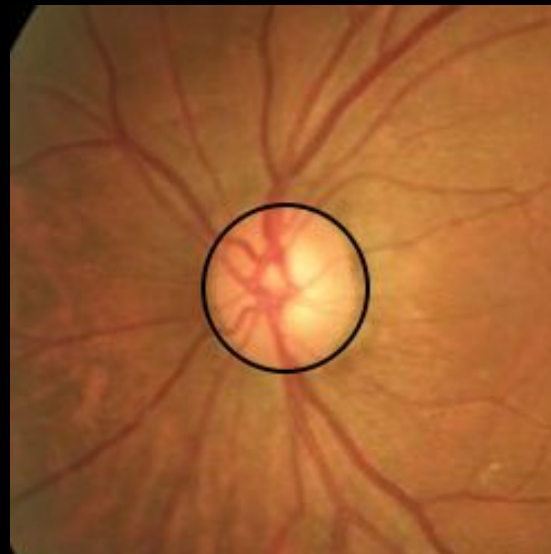
## 2. Ambiguous Ground-Truth, Imperfect Annotators

|                                                                                   |                             |                          |
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|  | Exchangeable annotators     | “Unequal” annotators     |
| $\nexists ! y^*$                                                                  | Aleatoric Image Uncertainty | <b>Annotation Styles</b> |
|                                                                                   |                             |                          |



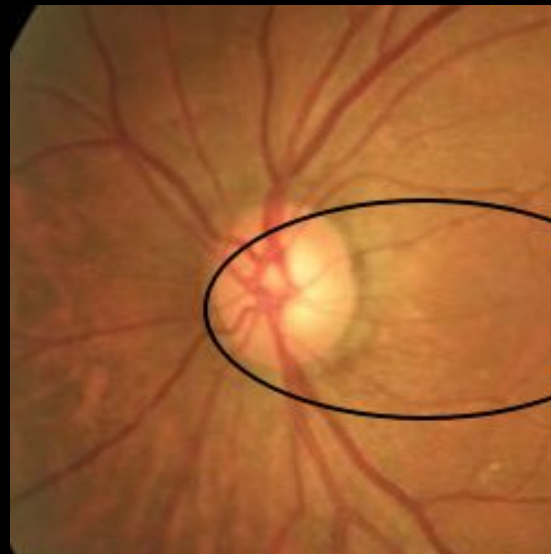
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|-----------------------------------------------------------------------------------|-----------------------------------------------|----------------------|
| $\nexists ! y^*$                                                                  | Aleatoric Image Uncertainty $\longrightarrow$ | Annotation Styles    |
|                                                                                   |                                               |                      |



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Post-doc

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Post-doc



About to graduate

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About to graduate



MSc student

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|                                                                                   |                                         |                             |
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| $\exists ! y^*$                                                                   | <b>Annotation Errors: “Label” Noise</b> | <b>Annotation Skills</b>    |

**Strategy: Keep/model disagreement. Avoid GT hard fusion.**

$\{1,1,1,0\} \rightarrow y=0.75$ , not majority 1!

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[aka rater-unweighted label smoothing]

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Fun Fact: Random label sampling is the same as this, iff you train with CE:

$$\text{BCE}(y,p) = -y \cdot \log(p) - (1-y) \cdot \log(1-p)$$

| y | prob. | CE(y,p)      |
|---|-------|--------------|
| 1 | 3/4   | $-\log(p)$   |
| 0 | 1/4   | $-\log(1-p)$ |

$$\mathbb{E}[\text{CE}(y,p)] = \underbrace{0.75 \cdot (-\log(p)) + 0.25 \cdot (-\log(1-p))}_{\text{CE}(y=0.75,p)}$$

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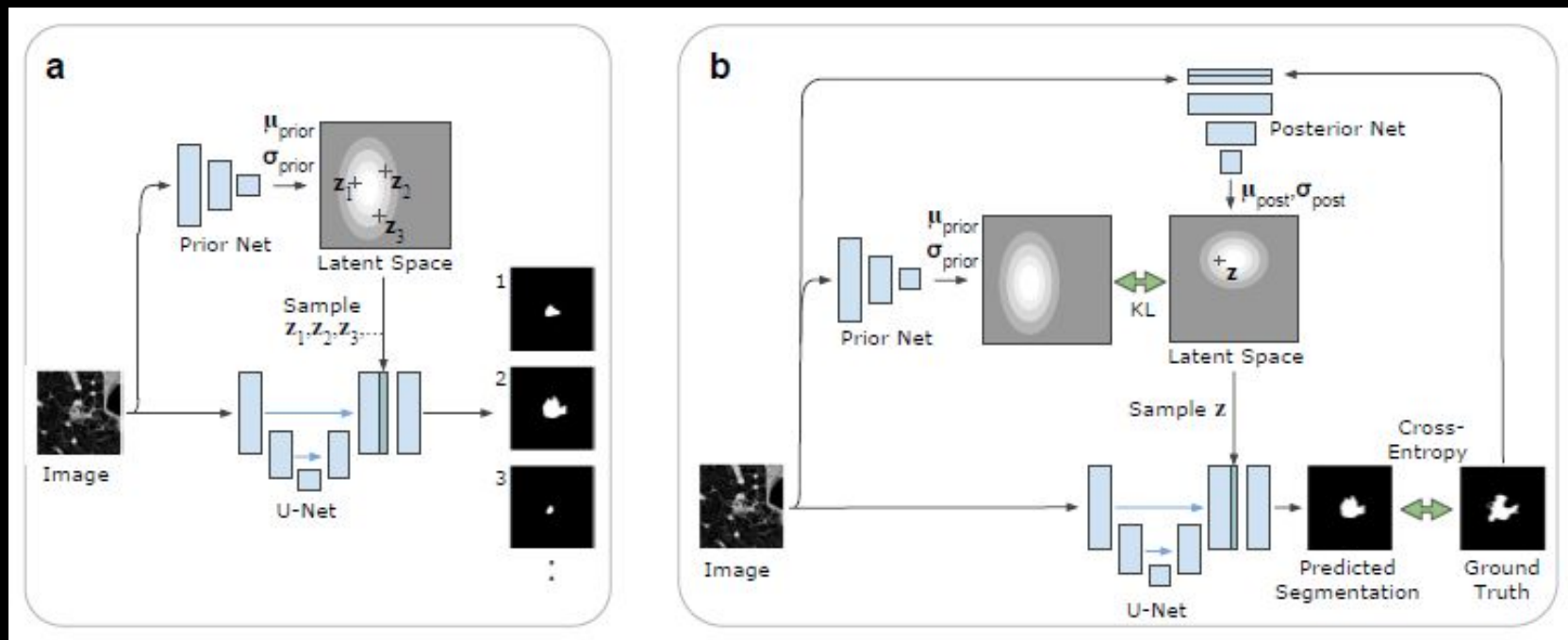
[aka rater-**unweighted** label smoothing]

**Fun Fact: Random label sampling is the same as this, iff you train with CE**

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**More advanced approaches: Probabilistic U-Net (Kohl NEURIPS 2018),  
PHiSeg (Baumgartner MICCAI 2019), and many more.**

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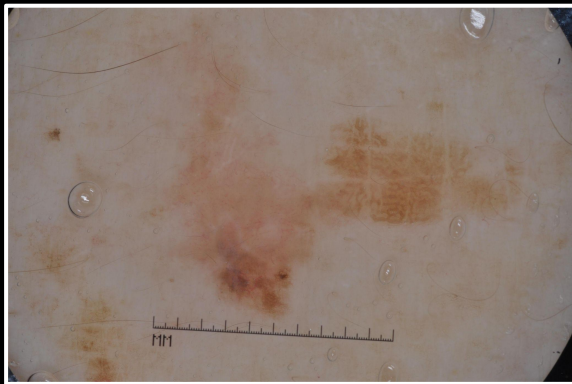


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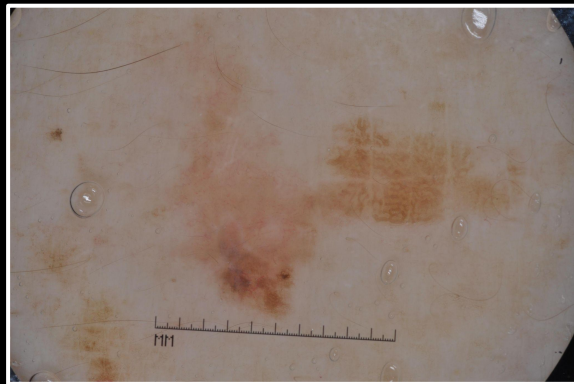
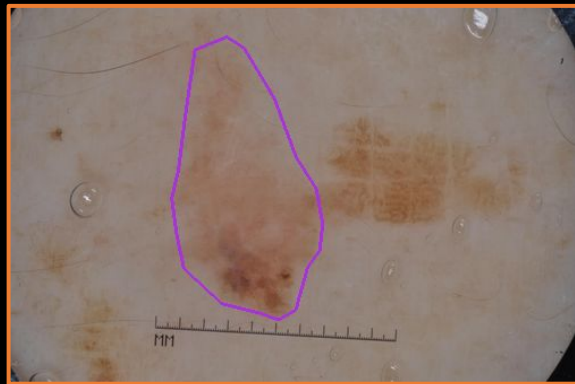
**Strategy: If possible discriminate IDs, train independent models**



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### **Further work:**

- Zepf et al., *That Label’s Got Style*, ICLR 2023: Prob. U-Net conditioned on tool
- Wu et al., *Diversified and Personalized Multi-rater Medical Image Segmentation*, CVPR 2024: Prob. U-Net conditioned on rater id
- Karimijafarbigloo et al., *Probabilistic Multi-rater Segmentation via Style-Aware Boundary Conditioning*, MICCAI 2026: disentangle shared content from rater-specific style
- Abhishek et al., *Segmentation Style Discovery: Application to Skin Lesion Images*, MICCAI ISIC 2024

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1st strategy: find outliers, merge, downweight, fix\*.

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\* Northcutt et al., Confident Learning: Estimating Uncertainty in Dataset Labels, JAIR 2021

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Or learn UQ !

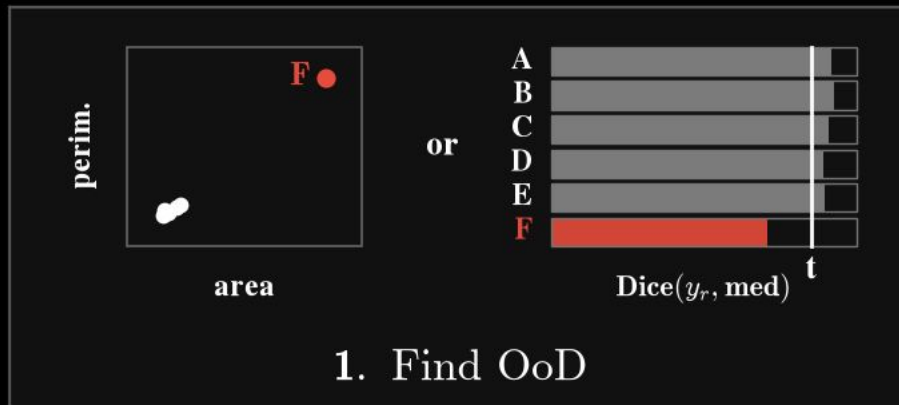
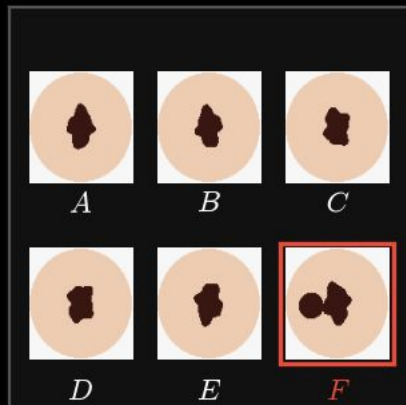


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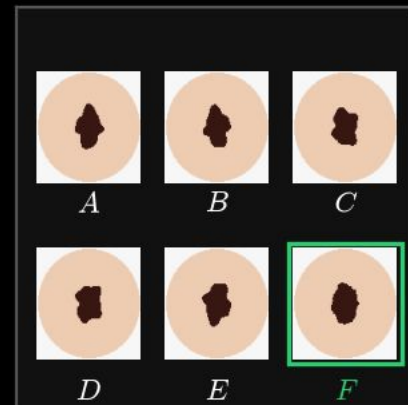
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2. Clean



## 2. Ambiguous Ground-Truth, Imperfect Annotators

|                                                                                   |                                                |                      |
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1st strategy: find outliers, merge, downweight, fix . 2nd: noisy label learning

Medical Image Analysis 65 (2020) 101759

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Deep learning with noisy labels: Exploring techniques and remedies in medical image analysis

Davood Karimi<sup>a,\*</sup>, Haoran Dou<sup>a</sup>, Simon K. Warfield<sup>a</sup>, Ali Gholipour<sup>a</sup>

*Department of Radiology, Boston Children's Hospital, Harvard Medical School, Boston, MA, USA*



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Strategy: try to model “annotation performance”.

### Simultaneous Truth and Performance Level Estimation (STAPLE): An Algorithm for the Validation of Image Segmentation

Simon K. Warfield<sup>\*</sup>, *Member, IEEE*, Kelly H. Zou, and William M. Wells, *Member, IEEE*

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|                                                                                   |                                  |                      |
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**Main Strategy: try to model “annotation performance”.**

*B. E-Step: Estimation of the Conditional Expectation of the Complete Data Log Likelihood Function*

In this section, the estimator for the unobserved true segmentation is derived. We first derive an expression for the conditional probability density function of the true segmentation given the expert decisions (observed segmentations) and the previous estimate of the performance parameters

$$\begin{aligned}
 f(\mathbf{T} | \mathbf{D}, \theta^{(k)}) &= \frac{f(\mathbf{D} | \mathbf{T}, \theta^{(k-1)}) f(\mathbf{T})}{\sum_{\mathbf{T}'} f(\mathbf{D} | \mathbf{T}', \theta^{(k-1)}) f(\mathbf{T}')} \\
 &= \frac{\prod_i [\prod_j f(D_{ij} | T_i, \theta_j^{(k-1)})] f(T_i)}{\sum_{T_1} \dots \sum_{T_n} \prod_i [\prod_j f(D_{ij} | T_i, \theta_j^{(k-1)})] f(T_i)} \\
 &= \frac{\prod_i [\prod_j f(D_{ij} | T_i, p_j, q_j^{(k-1)})] f(T_i)}{\prod_i [\sum_{T_i'} \prod_j f(D_{ij} | T_i', p_j, q_j^{(k-1)})] f(T_i')}
 \end{aligned}$$

such that at each voxel  $i$  we have

$$\begin{aligned}
 f(T_i | \mathbf{D}_i, \mathbf{p}^{(k-1)}, \mathbf{q}^{(k-1)}) &= \frac{\prod_j f(D_{ij} | T_i, p_j, q_j^{(k-1)}) f(T_i)}{\sum_{T_i'} \prod_j f(D_{ij} | T_i', p_j, q_j^{(k-1)}) f(T_i')} \quad (13)
 \end{aligned}$$

Considering (12), and noting that since  $\ln \prod_i f(T_i)$  is free of  $(\mathbf{p}, \mathbf{q})$  it follows that

$$(\mathbf{p}^{(k)}, \mathbf{q}^{(k)}) = \arg \max_{\mathbf{p}, \mathbf{q}} \sum_j \sum_i E \left[ \ln f(D_{ij} | T_i, p_j, q_j) | \mathbf{D}, \mathbf{p}^{(k-1)}, \mathbf{q}^{(k-1)} \right] \quad (17)$$

so for each expert  $j$

$$\begin{aligned}
 (p_j^{(k)}, q_j^{(k)}) &= \arg \max_{p_j, q_j} \sum_i E \left[ \ln f(D_{ij} | T_i, p_j, q_j) | \mathbf{D}, \mathbf{p}^{(k-1)}, \mathbf{q}^{(k-1)} \right] \\
 &= \arg \max_{p_j, q_j} \sum_i \left[ W_i^{(k-1)} \ln f(D_{ij} | T_i = 1, p_j, q_j) \right. \\
 &\quad \left. + (1 - W_i^{(k-1)}) \ln f(D_{ij} | T_i = 0, p_j, q_j) \right] \\
 &= \arg \max_{p_j, q_j} \sum_i W_i^{(k-1)} \ln p_j \\
 &\quad + \sum_{i: D_{ij}=1} (1 - W_i^{(k-1)}) \ln(1 - q_j) \\
 &\quad + \sum_{i: D_{ij}=0} W_i^{(k-1)} \ln(1 - p_j) \\
 &\quad + \sum_{i: D_{ij}=0} (1 - W_i^{(k-1)}) \ln q_j.
 \end{aligned}$$

$$\begin{aligned}
 \theta_j^{(k+1)} &= \arg \max_{\theta_j} \sum_i E \left[ \ln f(D_{ij} | T_i, \theta_j) | \mathbf{D}, \theta_j^{(k)} \right] \\
 &= \arg \max_{\theta_j} \sum_i \sum_s W_{is}^{(k)} \ln f(D_{ij} | T_i = s, \theta_j) \\
 &= \arg \max_{\theta_j} \sum_i \sum_{s': D_{ij}=s'} \sum_s W_{is}^{(k)} \\
 &\quad \times \ln f(D_{ij} = s' | T_i = s, \theta_j) \\
 &= \arg \max_{\theta_j} \sum_{s'} \sum_{i: D_{ij}=s'} \sum_s W_{is}^{(k)} \ln \theta_{js's}. \quad (21)
 \end{aligned}$$

Noting the constraint that each row of the rater parameter matrix must sum to one to be a probability mass function

$$\sum_{s'} \theta_{js's} = 1 \quad (22)$$

we can find the parameters that maximize the above expression by formulating the constrained optimization problem

$$\begin{aligned}
 0 &= \frac{\partial}{\partial \theta_{jn's}} \left[ \sum_{i'} \sum_{i: D_{ij}=i'} \sum_s W_{is}^{(k)} \ln \theta_{js's} \right. \\
 &\quad \left. + \lambda \sum_{s'} \theta_{js's} \right] \\
 &= \sum_{i: D_{ij}=i'} W_{is}^{(k)} \frac{1}{\theta_{jn's}} + \lambda.
 \end{aligned}$$

written in Gibbs form, as follows:

$$\begin{aligned}
 f(\mathbf{T}) &= \frac{1}{Z} \exp \left( \frac{-E(\mathbf{T})}{\tau} \right) \\
 Z &= \sum_{\mathbf{T}} \exp \left( \frac{-E(\mathbf{T})}{\tau} \right) \\
 -E(\mathbf{T}) &= \sum_i U(T_i) + \sum_i \sum_j V(T_i, T_j). \quad (25)
 \end{aligned}$$

This is the stationary Gibbs form, where  $Z$  and  $\tau$  are constants and  $E$  is called the energy function. Selecting  $E(\mathbf{T})$  as above, we can factor this standard form into a term depending solely on the single site voxel prior and a term depending on pairwise interactions

$$\begin{aligned}
 f(\mathbf{T}) &= \frac{1}{Z} \exp \left( \frac{\sum_i U(T_i)}{\tau} \right) \\
 &\quad \times \exp \left( \frac{\sum_i \sum_j V(T_i, T_j)}{\tau} \right) \\
 &= \frac{1}{Z_1} \exp \left( \frac{\sum_i U(T_i)}{\tau} \right) \frac{Z_1}{Z} \\
 &\quad \times \exp \left( \frac{\sum_i \sum_j V(T_i, T_j)}{\tau} \right) \\
 &= \prod_i f(T_i) \frac{Z_1}{Z} \exp \left( \frac{\sum_i \sum_j V(T_i, T_j)}{\tau} \right) \\
 \ln f(\mathbf{T}) &= \sum_i \ln f(T_i) + \ln Z_1 - \ln Z
 \end{aligned}$$

SEGMENTATION

909

$$\begin{aligned}
 &\cdot \left( \prod_j f(D_{ij} | T_i = 1, p_j^{(k)}, q_j^{(k)}) \right)^{T_i} \\
 &\cdot \left( \prod_j f(D_{ij} | T_i = 0, p_j^{(k)}, q_j^{(k)}) \right)^{1-T_i} \\
 &\propto \prod_i f(T_i) (q_i^{(k)})^{T_i} (p_i^{(k)})^{1-T_i}. \quad (29)
 \end{aligned}$$

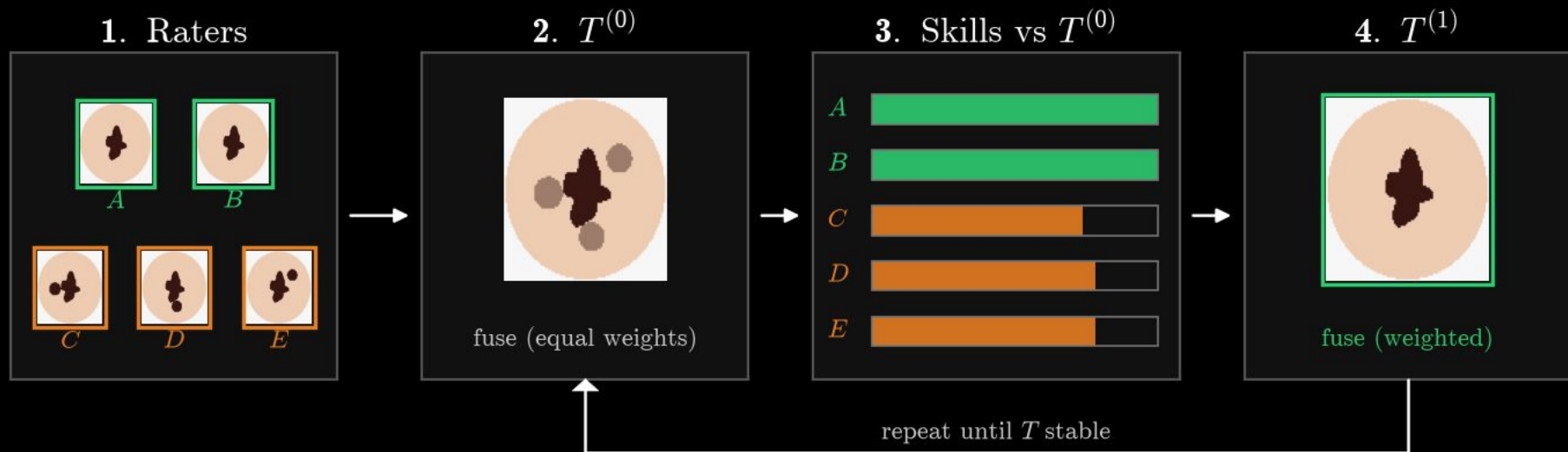
Taking the logarithm of this expression for the posterior probability and assuming a pairwise interaction neighborhood structure leads to

$$\begin{aligned}
 \ln f(\mathbf{T} | \mathbf{D}, \mathbf{p}^{(k)}, \mathbf{q}^{(k)}) &\propto \sum_{m=1}^N \sum_{n=1}^N \beta_{mn} (T_m T_n + (1 - T_m)(1 - T_n)) \\
 &\quad + \sum_{i=1}^N \ln \left( \frac{q_i^{(k)}}{p_i^{(k)}} \right) T_i \quad (30)
 \end{aligned}$$

where a term in (29) not depending on  $T_i$  has been ignored in (30) as it does not alter the estimate of the true segmentation, and  $\beta_{mn}$  is an interaction weight between voxels  $m$  and  $n$  that encodes the strength of the prior probability of true seg-

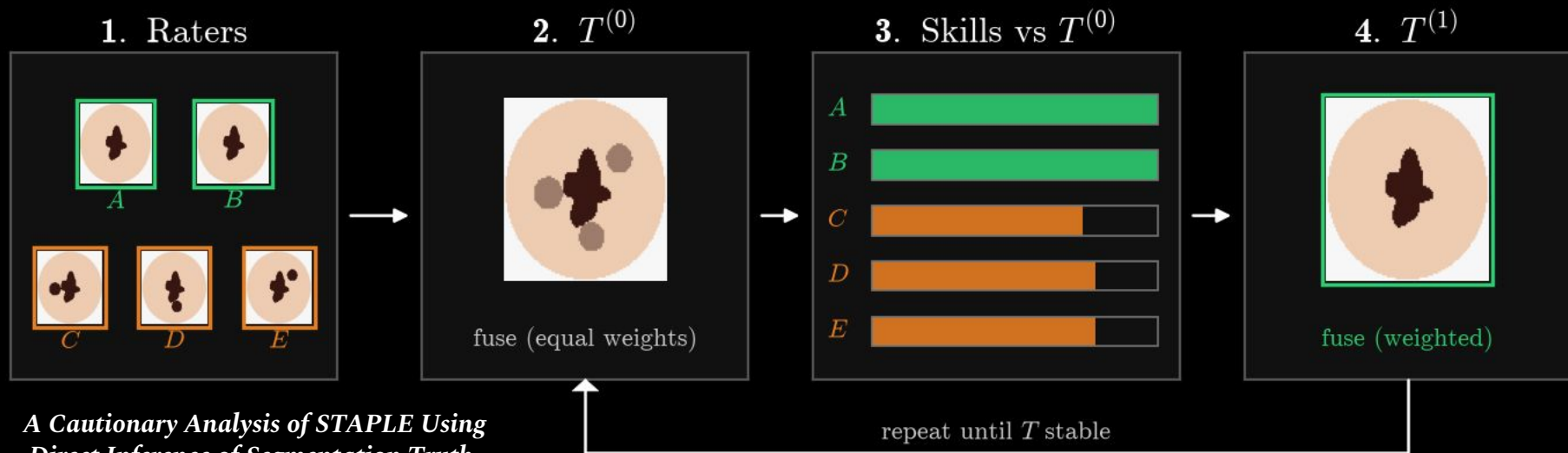
## 2. Ambiguous Ground-Truth, Imperfect Annotators

|                                                                                   |                                  |                      |
|-----------------------------------------------------------------------------------|----------------------------------|----------------------|
|  | Exchangeable annotators          | “Unequal” annotators |
| $\nexists ! y^*$                                                                  | Aleatoric Image Uncertainty      | Annotation Styles    |
| $\exists ! y^*$                                                                   | Annotation Errors: “Label” Noise | Annotation Skills    |



## 2. Ambiguous Ground-Truth, Imperfect Annotators

|                                                                                   |                                  |                      |
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*A Cautionary Analysis of STAPLE Using  
Direct Inference of Segmentation Truth,  
Van Leemput and Sabuncu, MICCAI 2014*

### 3. Evaluation and Multiple Raters

Model **validation** is as important as model training. Also quantifying inter-rater variability can tell us about reasonable performance limits ( $\# \text{ DSC}=1$ ).

**Advice:** consider building a **test set** with multiple annotations. It can be used to see if models have overfit to annotations or reflect proper uncertainty.

So suppose you do have a multi-annotator test set. What **metrics** can be used to see if we capture proper uncertainty?

# 3. Evaluation and Multiple Raters

|                                                                                   |                                    |                      |
|-----------------------------------------------------------------------------------|------------------------------------|----------------------|
|  | <b>Exchangeable annotators</b>     | “Unequal” annotators |
| $\nexists ! y^*$                                                                  | <b>Aleatoric Image Uncertainty</b> | Annotation Styles    |
| $\exists ! y^*$                                                                   | Annotation Errors: “Label” Noise   | Annotation Skills    |

**Strategy: Keep/model disagreement. Avoid GT hard fusion.**

---

**Use soft Dice against the soft consensus labels (~QUBIQ) or GED\* (Prob U-Net).**

Score against each ID. Report average and dispersion.

Clean labels, remove OoD, then score.

Score against expert(s), or against STAPLE.

\* GED: Sample multiple predictions, see if the distribution matches the distribution of the annotations. Rewards diversity.

### 3. Evaluation and Multiple Raters

|                                                                                   |                                  |                             |
|-----------------------------------------------------------------------------------|----------------------------------|-----------------------------|
|  | Exchangeable annotators          | <b>“Unequal” annotators</b> |
| $\nexists ! y^*$                                                                  | Aleatoric Image Uncertainty      | <b>Annotation Styles</b>    |
| $\exists ! y^*$                                                                   | Annotation Errors: “Label” Noise | Annotation Skills           |

**Strategy: If possible discriminate IDs, train independent models**

---

Use soft Dice against the soft consensus labels ( $\sim$ QUBIQ) or GED\* (Prob U-Net).

**Score against each ID. Report average and dispersion.**

Clean labels, remove OoD, then score.

Score against expert(s), or against STAPLE.

### 3. Evaluation and Multiple Raters

|                                                                                   |                                         |                      |
|-----------------------------------------------------------------------------------|-----------------------------------------|----------------------|
|  | <b>Exchangeable annotators</b>          | “Unequal” annotators |
| $\nexists ! y^*$                                                                  | Aleatoric Image Uncertainty             | Annotation Styles    |
| $\exists ! y^*$                                                                   | <b>Annotation Errors: “Label” Noise</b> | Annotation Skills    |

1st strategy: find outliers, merge, downweight, fix . 2nd: noisy label learning

---

Use soft Dice against the soft consensus labels ( $\sim$ QUBIQ) or GED\* (Prob U-Net).

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### 3. Evaluation and Multiple Raters

|                                                                                   |                                         |                      |
|-----------------------------------------------------------------------------------|-----------------------------------------|----------------------|
|  | <b>Exchangeable annotators</b>          | “Unequal” annotators |
| $\nexists ! y^*$                                                                  | Aleatoric Image Uncertainty             | Annotation Styles    |
| $\exists ! y^*$                                                                   | <b>Annotation Errors: “Label” Noise</b> | Annotation Skills    |

Strategy: try to model “annotation performance”.

---

Use soft Dice against the soft consensus labels ( $\sim$ QUBIQ) or GED\* (Prob U-Net).

Score against each ID. Report average and dispersion.

Clean labels, remove OoD, then score.

**Score against expert(s), or against STAPLE.**

## 4. Hands-On

Github repository:

<https://github.com/agaldran/uqinmia-miccai>

